Week #10 : Ecological Systems - Community Interactions

Subject/الموضوع: Pre-AP Biology Unit Title: Ecological Systems Grade9: الصف

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formative تكويني/ Summative ختامي / Differentiation التمايز	Resources and links for Parents المصادر و الروابط المخصصة لأولياء الأمور
Lesson 1: الحصة 1	I can explain how competition for limited resources shapes ecological communities. I can interpret data to identify resource overlap and niche partitioning.	1.1 Flexible display of information (size, color, font, layout).	Lesson 18: Competition, Niches & Resource Partitioning — investigation of habitat/time-use percentages, resource overlap and competitive exclusion. Teams / Pre-AP Biology.
Lesson 2 الحصة الثانية	I can use evidence to explain how niche partitioning reduces competition. I can predict effects of drought on resource availability, overlap and coexistence.	5.3 Build fluency with graduated scaffolds and models.	Lesson 18: Competition, Niches & Resource Partitioning — UAE desert drought cause–effect model, CER and AP-style questions. Teams / Pre-AP Biology.
Lesson 3 الحصة الثالثة	I can explain how predator and prey populations respond to each other. I can distinguish direct from indirect effects using population data.	6.3 Facilitate organization with graphic organizers, prompts, and note-taking guides.	Lesson 19: Predators, Keystone Species & Trophic Cascades — predator/herbivore/young-tree data investigation and trend modeling. Teams / Pre-AP Biology.
Lesson 4 الحصة الرابعة	I can explain the role of keystone species in community structure and diversity. I can model and predict trophic cascades across food webs.	2.1 Clarify vocabulary and symbols using visuals, connections, and supports.	Lesson 19: Predators, Keystone Species & Trophic Cascades — keystone-species models, UAE desert cascade scenario and AP-style reasoning. Teams / Pre-AP Biology.
Lesson 5 الحصة الخامسة	I can identify mutualism, parasitism and commensalism from evidence. I can explain how symbiosis changes environmental pressures for one or both partners.	8.2 Vary challenge level with differentiated tools, resources, and difficulty.	Lesson 20: Symbiosis — evidence classification (+/+, +/0, +/−), UAE coral reef cleaner-fish model and AP-style questions. Teams / Pre-AP Biology.

Teacher's notes/ Reminders **ملاحظات المعلم والتذكير**	• Bring Biology notebook for data analysis, cause–effect diagrams, CER writing and AP-style responses. • Review lesson PowerPoints and answer-key/peer-feedback slides on Teams. • Emphasize evidence-based claims and distinguish correlation from causation.

الأسبوع# Week #10

Technology Integration دمج التكنولوجيا	• PowerPoint ecological models, tables, diagrams and answer-key/peer-feedback slides. • Microsoft Teams for sharing/reviewing learning resources.
Cross-curricular links الروابط بين المواد الدراسية	• Mathematics: percentages, trends and interpreting population tables. • Geography/Environmental Science: UAE desert drought, resource scarcity and coastal ecosystems. • Literacy: evidence-based CER arguments and annotated cause–effect models.
External Assessment Integration دمج التقييمات الخارجية	Pre-AP Biology Unit 1: Ecological Systems; ECO 4 Community Interactions. AP-style data interpretation, causal modeling and justification from evidence.
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	• Thinking: pattern recognition, systems thinking, classification and causal reasoning. • Communication: evidence-based explanations, CER, peer feedback and model annotation. • Collaboration: pair investigations, case analyses and poster/model discussion.
Standards المعايير	• Lessons 1–2: ECO 4.1(a–b), EK 4.1.1c–d — competition and niche partitioning. • Lessons 3–4: ECO 4.1(c), EK 4.1.1a–b — predator–prey dynamics, keystone species and trophic cascades. • Lesson 5: ECO 4.2(a–b), EK 4.2.1a–c — mutualism, parasitism and commensalism.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	• Lesson 1: How can similar species coexist without one eliminating the other? Lesson 2: How can drought change resource overlap and competition? Lesson 3: How can predator changes indirectly affect plants? Lesson 4: How can one keystone species reshape a whole community? Lesson 5: How can close relationships reduce pressures—or create costs?
Success Criteria معايير النجاح **(Measurable)**	Lesson 1 — Competition & Niche Overlap MUST: Identify competition and niche partitioning. SHOULD: Explain how resource overlap increases competition. COULD: Analyze space/time-use data to justify coexistence. Lesson 2 — Resource Partitioning & UAE Drought MUST: Identify shared limited resources. SHOULD: Model drought → overlap → competition. COULD: Predict different outcomes for flexible versus inflexible species. Lesson 3 — Predator–Prey Dynamics MUST: Identify predators, prey and direct/indirect effects. SHOULD: Explain population trends using data. COULD: Assess what further evidence would test a causal claim.

	Lesson 4 — Keystone Species & Trophic Cascades MUST: Define keystone species and trophic cascade. SHOULD: Explain direct and indirect effects of predator change. COULD: Predict a community-wide cascade and limitations of the model. Lesson 5 — Symbiosis MUST: Classify mutualism, parasitism and commensalism. SHOULD: Explain benefits, harms or no measurable effects for both partners. COULD: Justify effects of partner loss using evidence.
National Identity Domain **مجالات الربط بالهوية الوطنية**	Conservation الحفاظ علي البيئة Respect الاحترام Heritage التراث
	المجال Area: Conservation / Sustainability / Community
	• UAE deserts: investigate how species partition scarce food, water, space and activity times. • UAE protected reserves: model drought-driven competition and trophic cascades involving predators, herbivores and shrubs. • UAE coral reefs: examine cleaner-fish mutualism and effects of disturbance on community relationships.